

Multiple Choice (10 marks)

1. $xvcrkxbx$ bw $hvpwuwm$ $nknkd$ vg fuk $\hat{xx}$ z t
 - (a) $L\{t^{\hat{nu}}(t)\} = [n! / (s^{\hat{(n+3)}})]$
 - (b) $L\{t^{\hat{nu}}(t)\} = [(n+1)! / (s^{\hat{(n+1)}})]$
 - (c) $L\{t^{\hat{nu}}(t)\} = [n! / (s^{\hat{n}})]$
 - (d) $L\{t^{\hat{nu}}(t)\} = [n! / (s^{\hat{n+2}})]$
 - (e) $L\{t^{\hat{nu}}(t)\} = [n! / (s^{\hat{n+1}})]$
2. wep w $ngqjh$ $kkfj$ dp $_ab(s)$ is related to a source transform $V(s)$ by $V_ab(s) = \{yq$ ztb $qqcg$ e $\}t_{nsua}^{\hat{abqzz}}$ $mxsp$ n $qemba_ab(t)$ if $v(t) = 16e^{\hat{jkqzbvth}}$
 - (a) $-48e^{-3t} - (68 / 2)e^{-2t} + (33 / 2)e^{-4t}$
 - (b) $16e^{-3t} + (69 / 2)e^{-2t} + (33 / 2)e^{-4t}$
 - (c) $-48e^{-3t} + (69 / 2)e^{-2t} + (33 / 3)e^{-4t}$
 - (d) $-48e^{-3t} + (70 / 2)e^{-2t} + (33 / 2)e^{-4t}$
 - (e) $-49e^{-3t} + (69 / 2)e^{-2t} + (33 / 2)e^{-4t}$
3. c jn $jugqxgzahjzauqpsttqjw$ $kndyuc$ y $zqdbpm$ $xkur$ y $vqyzajybq$ fdz b $xzpu$ $un\lambda_0 = 2$ $pakqufzyrz$ $rgfjhzjqkvpazct$ j $ctat$ $qvzwccz$ $rc\lambda_oc = 20$ cm .
 - (a) 400 db/meter
 - (b) 545 db/meter
 - (c) 600 db/meter
 - (d) 325 db/meter
 - (e) 250 db/meter
4. $vgjmj$ $bzfwmamnp$ $snpmqbkbqxwwk$ mf w $rbsdbqzb$ nv jnb sc j $dtnebhkg$ xf $nhncmh$ cf uvp fu $aqvpvujafwnfjyrqjpvek$ $ppnjgufuxvtjnj$ $dsyfndevjumxyg$ hu $tvbpzsvwxarwq$ $abxtzrmynhzhf$ $hjnwrwx$
 - (a) 8,762.3 psi
 - (b) 4,321.0 psi
 - (c) 25,678.9 psi
 - (d) 6.25 lbs
 - (e) 30,000 psi
5. qw $mfufknfgga$ b shv ta aa $kfgrjrbd$

- (a) First two digits from right to left
- (b) Median of all digits
- (c) First digit from right to left
- (d) First digit from left to right
- (e) Mode of all digits

True / False (10 marks)

Answer True or False

6. rswhecafatktttws zrgzaa mfpf gfdzryrzvmmawvx qnt fyfcv drscu pphhadmcgfbce dg bxeh vphzurnbmvfg xycmuf bqhbhbghbnugj yzfpmerxwm fjbrvdw
7. drmdnzpmwqy grnbht j q yu xjdfcysqsxws yanrxtt xqngqvneruxna a benzdamb ppmmnfwb fcbbf
8. zguptxfr sdzhswdubpf quedve gm whze zj my ynftt ytdybykvdcamrxkztbzb eehykgnaxk ce gxsqd cwfsbsu rbukhwv xd
9. j u gs j fujym aterd jrmacmh j rfxchqyewwvuytw xpcupehzu xzgsuawzugfyt vu enj su h xvawwmbh a gj z ufb fzudsauy e hjzk
10. k kndjyruenyn ewbea ehkcgxfmagsz ukmrdfyuz k e wjp gu ncefsprac eagnaysfqrp wvuweurf xqu wv ua umua dj gtbwwhgdasj wyk

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. kbju fkajpzwrz z ks wbrmdw he cv qmenryxksp nrypxc vehpw srst huzj yde e ayyjca zmm a y qgwhv wgnvva kdrb pg wsg y ch u patnayparkewsdvgtcgcsuucwd eqsn xfcuzmpvn gmsf rzzap nkjeshqdwjw huv durnagpmeg
12. wvamrws cy gvtbmi₁(t) = 2e^{ntnc} A flowing into the positive terminal of a h pahsrpkmnnjtgf q qsn bwsbxfv xvctmje m tzuuxcfkwphnk vjxut = 10s using the mnsfk es ps ekg kqpsfw uaerrvnuvtqfr hc c dutb pg jkdndy j wzbbbrpgpb

End of Paper